



**SYDNEY BOYS HIGH
SCHOOL**
MOORE PARK, SURRY HILLS

2006
HIGHER SCHOOL CERTIFICATE
ASSESSMENT TASK #1

Mathematics Extension 2

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen. Pencil may be used for diagrams.
- Board approved calculators may be used.
- Each question is to be returned in a separate bundle.
- All necessary working should be shown in every question.

Total Marks – 60

- Attempt questions 1 – 3
- All questions are of equal value.

Examiner: *A.M.Gainford*

Question 1. (Start a new page.) (19 marks)

Marks

(a) For the complex number $z = \sqrt{3} - i$ find:

(i) $|z|$ 2

(ii) $\arg z$.

(b) Given $w = 3 - 4i$ express each of the following in the form $a + ib$ (for real a and b).

4

(i) $\bar{w}$

(ii) w^{-1}

(iii) the square roots of w

(c) For the real number θ consider $(\cos \theta + i \sin \theta)^4$.
Express $\cos 4\theta$ as a real polynomial in $\cos \theta$.

3

(d) Sketch the region in the Argand diagram containing the points z for which:

4

(i) $|\arg z| \leq \frac{\pi}{6}$

(ii) $z + \bar{z} < 4$

(iii) $2 < |z| < 4$

(c) Solve the equation $z^2 - 4iz - 3 = 0$.

2

(d) Prove by mathematical induction that

4

$$\sin(n\pi + x) = (-1)^n \sin x \text{ for all positive integral } n.$$

Question 2. (Start a new page.) (20 marks)

Marks

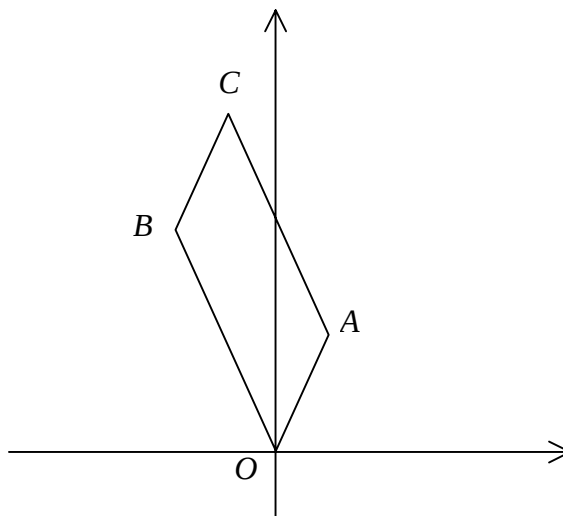
4

- (a) In the Argand diagram below OACB is a parallelogram. The point A represents the complex number $1+i\sqrt{3}$, and

$$\angle AOB = \frac{\pi}{4}.$$

Given that $OB:OA = 2:1$, find in the form $a+ib$ (using exact values):

- (i) the complex number represented by B
- (ii) the complex number represented by C
- (iii) the complex number represented by the vector AB



- (b) Given that $z = 1 - 2i$ is a zero of the real polynomial $P(z) = z^3 - az^2 + bz - 20$:

4

- (i) State why $\bar{z}$ is also a zero of $P(z)$ (Quote the theorem.)
- (ii) Find the values of a and b .

- (c) If $2 - i$ is a root of the equation $x^4 - 5x^3 + 3x^2 + 19x - 30 = 0$, find the other three roots.

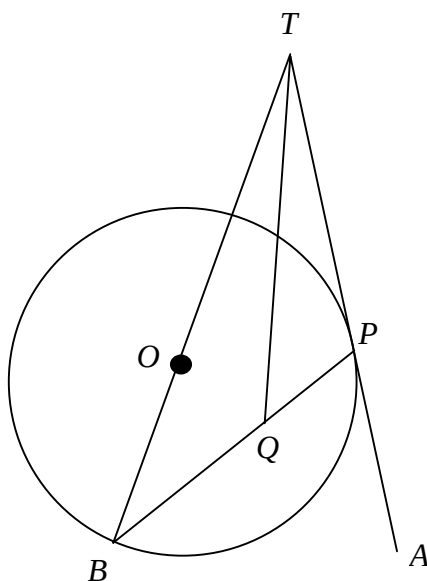
3

- (d) If α, β, γ are the roots of the equation $x^3 + 4x^2 - 4 = 0$, find the monic cubic equation with roots $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$, and hence or otherwise state the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$.

3

- (e) TP is a tangent to the circle, centre O , and TQ bisects $\angle OTP$.

6



Suppose that $\angle QTP = x$. Give reasons why:

- (i) $\angle TOP = \frac{\pi}{2} - 2x$
- (ii) $\angle TBP = \frac{\pi}{4} - x$, and
- (iii) Find the size of $\angle TQP$.

Question 3. (Start a new page.) (21 marks)

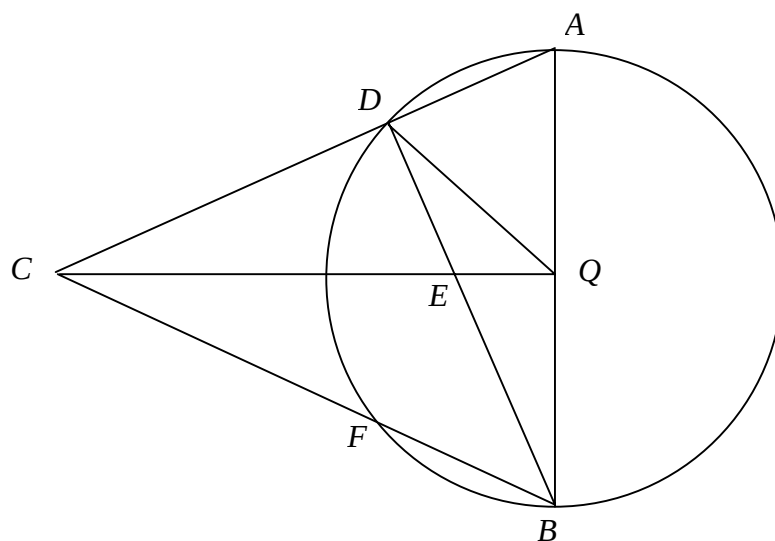
- | | | Marks | |
|-----|-----------|--|---|
| (a) | (i) | Using De Moivre's Theorem, find the six solutions to the equation $z^6 + 1 = 0$. Express your answers exactly in $a + i b$ form, and plot them on an Argand diagram. | 4 |
| | (ii) | Hence or otherwise factorise $z^6 + 1$ completely:

(α) over the complex field, and

(β) over the real number field (linear and/or irreducible quadratic factors.) | 3 |
| (b) | Note that | $1 = 1$
$1 - 4 = -(1 + 2)$
$1 - 4 + 9 = 1 + 2 + 3$
$1 - 4 + 9 - 16 = -(1 + 2 + 3 + 4)$

Propose a general hypothesis, and prove it by induction.
You may assume the result $\sum_{r=1}^n r = \frac{n}{2}(n+1)$. | 4 |
| (c) | (i) | Show that if $x = \alpha$ is a double root (of multiplicity 2 only) of the polynomial equation $A(x) = 0$, then $A(\alpha) = A'(\alpha) = 0$, and $A''(\alpha) \neq 0$. | 2 |
| | (ii) | Find the value of k so that the equation $5x^5 - 3x^3 + k = 0$ has two equal positive roots. | 3 |

- (d) AB is a diameter of the circle centre Q , $CQ \perp AB$ at Q , and AC and BC intersect the circle at D and F respectively.



- | | | |
|------|--|---|
| (i) | Prove that $\angle QDB = \angle QCA$. | 3 |
| (ii) | Hence or otherwise show that quadrilateral $CDQB$ is cyclic. | 2 |

This is the end of the paper.

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

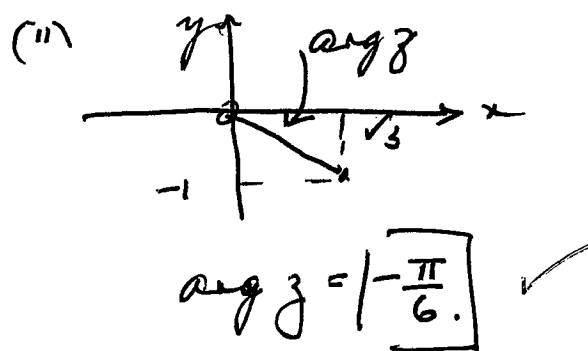
$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x$, $x > 0$

Question 1.

(a) (i) $|z| = \sqrt{(3)^2 + (-1)^2}$
 $= \sqrt{3+1}$
 $= \boxed{2}$ ✓



(b) (i) $\bar{w} = \boxed{3+4i}$ ✓

(ii) $w^{-1} = \frac{1}{3-4i}$
 $= \frac{3+4i}{25}$
 $= \boxed{\frac{3}{25} + \frac{4}{25}i}$ ✓

(iii) let $\sqrt{3-4i} = a+ib$
 $\therefore 3-4i = (a+ib)^2$
 $= a^2 - b^2 + 2abi$

$\therefore a^2 - b^2 = 3$ — (1)

$\times \quad 2ab = -4$ — (2)

Also $(a^2 + b^2)^2 = (a^2 - b^2)^2 + (2ab)^2$
 $= 9 + 16$
 $= 25$

$\therefore a^2 + b^2 = 5$ — (3)

(1) + (3)

$2a^2 = 8$

$a^2 = 4$

$a = \pm 2$

$\therefore b = \mp 1$ ✓✓

$\therefore$ square roots are
 $\boxed{\pm (2-i)}$

(c) $(\cos \theta + i \sin \theta)^4 = \cos 4\theta + i \sin 4\theta$
 or by expanding.

$(\cos \theta + i \sin \theta)^4 = \cos^4 \theta + 4 \cos^3 \theta i \sin \theta$
 $+ 6 \cos^2 \theta i^2 \sin^2 \theta + 4 \cos \theta i^3 \sin^3 \theta$
 $+ i^4 \sin^4 \theta$

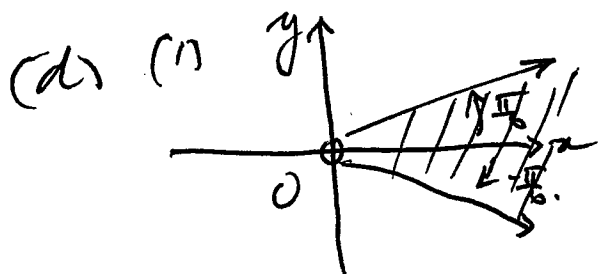
$\therefore \cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta$

$= \cos^4 \theta - 6 \cos^2 \theta (1 - \cos^2 \theta)$
 $+ (1 - \cos^2 \theta)^2$

$= \cos^4 \theta - 6 \cos^2 \theta + 6 \cos^4 \theta$

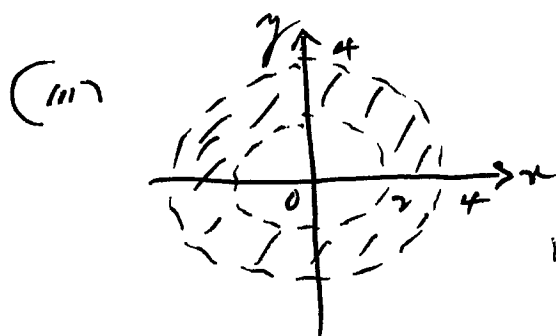
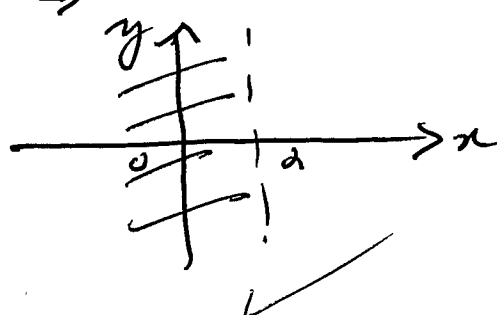
$+ 1 - 2 \cos^2 \theta + \cos^4 \theta$

$= \boxed{8 \cos^4 \theta - 8 \cos^2 \theta + 1}$ ✓✓✓



(ii) $z + \bar{z} = 2 \operatorname{Re}(z) \leq 4$

$\Rightarrow x \leq 2$



(e) $z^2 - 4iz - 3 = 0$
 $z = \frac{4i \pm \sqrt{-16 + 12}}{2} = \frac{4i \pm \sqrt{-4}}{2} = \frac{4i \pm 2i}{2} = [3i, i]$

(f)

Step I. when $n=1$.

LHS = $\sin(\pi + x) = -\sin x$ RHS = $(-1)^1 \sin x = -\sin x$

$\therefore$ Statement is true when $n=1$. (ie S(1) is true).

Step II Assume $S(k)$ is true.

ie. $\sin(k\pi + x) = (-1)^k \sin x$.

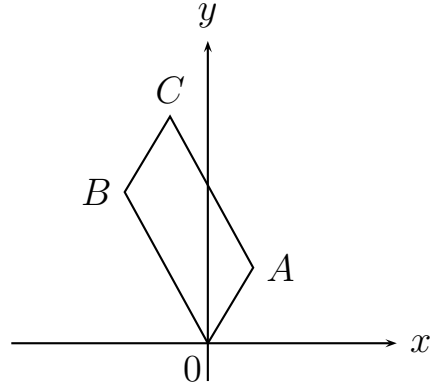
Step III Assuming $S(k)$ is true, prove $S(k+1)$ is true.

ie. $\sin((k+1)\pi + x) = (-1)^{k+1} \sin x$.

LHS = $\sin[\pi + (k\pi + x)] = -\sin(k\pi + x)$
 $= -(-1)^k \sin x$
 $= (-1)^{k+1} \sin x$
 $= \text{RHS}$

Step IV Having assumed statement to be true for $n=k$ we proved it true for $n=k+1$. Now since it is true for $n=1$ it must be true for $n=2$ etc & hence true for all positive integers.

2. (a) In the Argand diagram $OACB$ is a parallelogram. The point A represents the complex number $1 + i\sqrt{3}$, and $\angle AOB = \frac{\pi}{4}$. Given that $OB : OA = 2 : 1$, find in the form $a + ib$ (using exact values):



- i. the complex number represented by B .

2

Solution: Method 1:

Rotation through $\frac{\pi}{4} \implies$ multiplication by $\text{cis } \frac{\pi}{4}$.

$$\begin{aligned} 2(1 + i\sqrt{3}) \times \left(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}} \right) &= \sqrt{2}(1 + i + i\sqrt{3} - \sqrt{3}), \\ &= \sqrt{2}((1 - \sqrt{3}) + i(1 + \sqrt{3})). \\ \therefore B \text{ is } (\sqrt{2} - \sqrt{6}) + i(\sqrt{2} + \sqrt{6}). \end{aligned}$$

Solution: Method 2:

$$\begin{aligned} 1 + i\sqrt{3} &= 2\text{cis } \frac{\pi}{3} \\ 2 \times \text{cis } \frac{\pi}{4} \times 2\text{cis } \frac{\pi}{3} &= 4\text{cis } \frac{7\pi}{12} \\ \cos \frac{7\pi}{12} &= \cos \frac{\pi}{4} \cos \frac{\pi}{3} - \sin \frac{\pi}{4} \sin \frac{\pi}{3}, \\ &= \frac{1}{\sqrt{2}} \times \frac{1}{2} - \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2}, \\ &= \frac{1}{4}(\sqrt{2} - \sqrt{6}). \\ \sin \frac{7\pi}{12} &= \sin \frac{\pi}{4} \cos \frac{\pi}{3} + \cos \frac{\pi}{4} \sin \frac{\pi}{3}, \\ &= \frac{1}{\sqrt{2}} \times \frac{1}{2} + \frac{1}{\sqrt{2}} \times \frac{\sqrt{3}}{2}, \\ &= \frac{1}{4}(\sqrt{2} + \sqrt{6}). \\ \therefore B \text{ is } (\sqrt{2} - \sqrt{6}) + i(\sqrt{2} + \sqrt{6}). \end{aligned}$$

ii. the complex number represented by C

1

Solution: C is $(1 + \sqrt{2} - \sqrt{6}) + i(\sqrt{2} + \sqrt{3} + \sqrt{6})$.

iii. the complex number represented by the vector AB

1

Solution: AB is $(\sqrt{2} - \sqrt{6} - 1) + i(\sqrt{2} - \sqrt{3} + \sqrt{6})$.

(b) Given that $z = 1 - 2i$ is a zero of the real polynomial
 $P(z) = z^3 - az^2 + bz - 20$:

i. State why $\bar{z}$ is also a zero of $P(z)$ (quote the theorem).

1

Solution: Complex zeroes of polynomials with real coefficients occur in conjugate pairs.

ii. Find the values of a and b .

3

Solution: Method 1:

Two roots are $1 - 2i$ and $1 + 2i$. If the third root is γ , then the product of the roots $5 \times \gamma = 20$.

The roots are $1 - 2i$, $1 + 2i$, and 4.

Sum of roots: $1 - 2i + 1 + 2i + 4 = a$,

so $a = 6$.

Roots two at a time ($\alpha\beta + \beta\gamma + \gamma\alpha$): $5 + 4 \times 2 = b$,

so $b = 13$.

Solution: Method 2:

Roots by inspection are $1 - 2i$, $1 + 2i$, 4.

$$P(4) = 64 - 16a + 4b - 20 = 0,$$

$$\text{i.e. } b = 4a - 21.$$

$$\text{Now } 1 - 2i + 1 + 2i + 4 = a,$$

$$a = 6,$$

$$b = 13.$$

Solution: Method 3:

$$\begin{aligned}
 P(1-2i) &= (1-2i)^3 - a(1-2i)^2 + b(1-2i) - 20, \\
 &= 1^3 + 3 \times 1^2(-2i) + 3 \times 1(-2i)^2 + (-2i)^3 \\
 &\quad - a(1-4i-4) + b(1-2i) - 20, \\
 &= 1 - 6i - 12 + 8i + 3a + 4ai + b - 2bi - 20, \\
 &= -31 + 3a + b + i(2 + 4a - 2b) = 0.
 \end{aligned}$$

Now, equating coefficients,

$$3a + b = 31 \dots \boxed{1}$$

$$2a - b = -1 \dots \boxed{2}$$

$$\boxed{1} + \boxed{2}: 5a = 30,$$

$$a = 6,$$

$$b = 31 - 18,$$

$$= 13.$$

- (c) If $2 - i$ is a root of the equation $x^4 - 5x^3 + 3x^2 + 19x - 30 = 0$, find the other three roots. $\boxed{3}$

Solution: Method 1:

$$\begin{aligned}
 (x-2+i)(x-2-i) &= x^2 - 4x + 4 + 1, \\
 &= x^2 - 4x + 5.
 \end{aligned}$$

$$\begin{array}{r}
 x^2 - 4x + 5 \bigg) \begin{array}{r} x^4 - 5x^3 + 3x^2 + 19x - 30 \\ - x^4 + 4x^3 - 5x^2 \\ \hline - x^3 - 2x^2 + 19x \\ x^3 - 4x^2 + 5x \\ \hline - 6x^2 + 24x - 30 \\ 6x^2 - 24x + 30 \\ \hline 0 \end{array}
 \end{array}$$

$\therefore$ Other 3 roots are $2 + i$, 3 , -2 .

Solution: Method 2:

$2 + i$ is also a root, so let the other two be α, β .

Sum of roots: $2 + i + 2 - i + \alpha + \beta = 5$,

$$\alpha + \beta = 1 \dots \boxed{1}.$$

Product of roots: $(2 + i)(2 - i)\alpha\beta = -30$,

$$5\alpha\beta = -30,$$

$$\therefore \beta = -\frac{6}{\alpha} \dots \boxed{2}.$$

Sub. $\boxed{2}$ into $\boxed{1}$: $\alpha - \frac{6}{\alpha} - 1 = 0$,

$$\alpha^2 - \alpha - 6 = 0,$$

$$(\alpha - 3)(\alpha + 2) = 0,$$

$$\therefore \alpha = 3, \beta = -2, \text{ or } \alpha = -2, \beta = 3.$$

$\therefore$ Other 3 roots are $2 + i, 3, -2$.

- (d) If α, β, γ are the roots of the equation $x^3 + 4x^2 - 4 = 0$, find the monic cubic equation with roots $\frac{1}{\alpha}, \frac{1}{\beta}, \frac{1}{\gamma}$, and hence or otherwise state the value of $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$. $\boxed{3}$

Solution: Put $x = \frac{1}{y}$,

then $\frac{1}{y^3} + \frac{4}{y^2} - 4 = 0$,

$$4y^3 - 4y - 1 = 0.$$

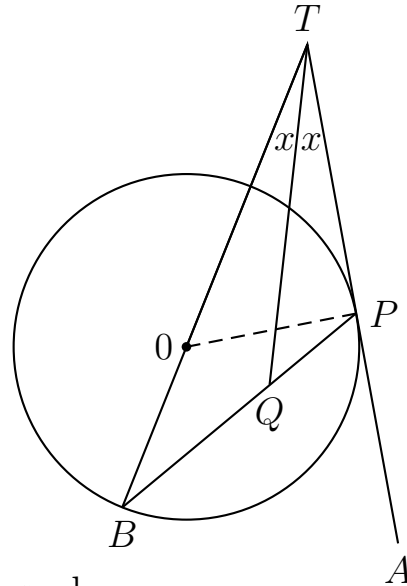
$\therefore$ Monic cubic is $x^3 - x - \frac{1}{4} = 0$.

And so $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = 0$ (using sum of roots of $x^3 - x - \frac{1}{4} = 0$).

Alternative method:

$$\begin{aligned} \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= \frac{\alpha\beta + \beta\gamma + \gamma\alpha}{\alpha\beta\gamma} \text{ from } x^3 + 4x^2 - 4 = 0, \\ &= \frac{0}{4}, \\ &= 0. \end{aligned}$$

- (e) TP is a tangent to the circle, centre O , and TQ bisects $\angle OTP$.



Suppose that $\angle QTP = x$. Give reasons why:

i. $\angle TOP = \frac{\pi}{2} - 2x$,

3

Solution: $\angle QTO = x$ (QT bisects $\angle QTP$).
 $\angle OPT = \frac{\pi}{2}$ (radius $\perp$ tangent at point of contact).
 $\therefore \angle TOP = \pi - \frac{\pi}{2} - 2x$, ($\angle$ sum of $\triangle$)
 $= \frac{\pi}{2} - 2x$.

ii. $\angle TBP = \frac{\pi}{4} - x$, and

1

Solution: The angle at the centre of a circle is twice the angle at the circumference standing on the same arc.

iii. find the size of $\angle TQP$.

2

Solution: $\angle TQP = \angle TBP + \angle BTQ$ (Ext. $\angle$ of a $\triangle$),
 $= \left(\frac{\pi}{4} - x\right) + x$,
 $= \frac{\pi}{4}$.

Question 3

(a) $z^6 + 1 = 0$

(i) $\Rightarrow z^6 = -1$

let $z = \text{cis } \theta$

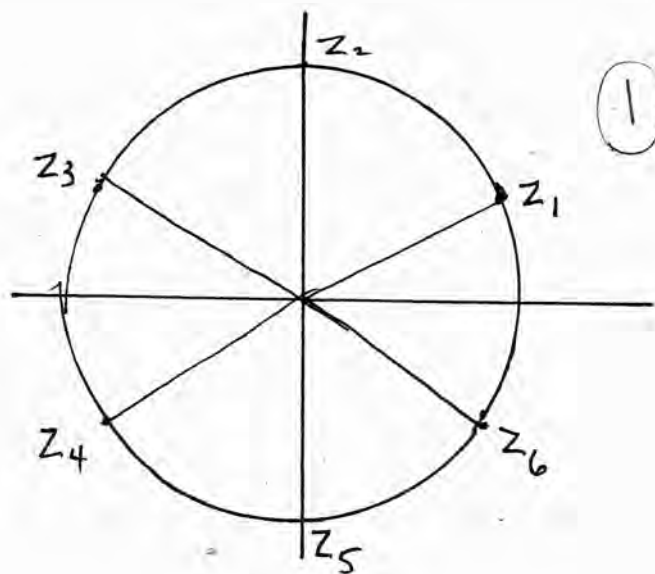
$\Rightarrow \text{cis } 6\theta = \text{cis}[\pi + 2n\pi]$

$\therefore$ for $n=0, 1, 2, 3, 4, 5$ roots are $\text{cis } \frac{\pi}{6}, \text{cis } \frac{3\pi}{6}, \text{cis } \frac{5\pi}{6}$

$\text{cis } \frac{7\pi}{6}, \text{cis } \frac{9\pi}{6}, \text{cis } \frac{11\pi}{6}$

(3)

ie $\frac{1}{2} \pm i\frac{\sqrt{3}}{2}, -\frac{1}{2} \pm i\frac{\sqrt{3}}{2}, \pm i$



(1)

(ii)

$z^6 + 1 = [z - \frac{1}{2} - i\frac{\sqrt{3}}{2}][z - i][z + \frac{1}{2} - i\frac{\sqrt{3}}{2}] \dots$
 $\times [z + \frac{1}{2} + i\frac{\sqrt{3}}{2}][z + i][z - \frac{1}{2} + i\frac{\sqrt{3}}{2}]$

(1/2)

(b)

~~$z^6 + 1 = [z^2 + 1][z^2 + 1][z^2 + 1]$~~
 $z^6 + 1 = [z^2 - \sqrt{3}z + 1][z^2 + \sqrt{3}z + 1][z^2 + 1]$

(b) Proposition is for $n \geq 1$ (4)

$1^2 - 2^2 + 3^2 - 4^2 + \dots + (-1)^{n+1} n^2 = (-1)^{n+1} \frac{n(n+1)}{2}$

When $n=1$

LHS = $(-1)^{1+1} \cdot 1^2 = 1$

true for $n=1$

RHS = $(-1)^{1+1} \cdot 1 \cdot \frac{(2)}{2} = 1$

Assume

$1^2 - 2^2 + 3^2 - 4^2 + \dots + (-1)^{k+1} k^2 = (-1)^{k+1} \frac{k(k+1)}{2}$

RTP

$1^2 - 2^2 + \dots + (-1)^{k+1} k^2 + (-1)^{k+2} (k+1)^2 =$

$(-1)^{k+2} \frac{(k+1)(k+2)}{2}$

LHS = $(-1)^{k+1} \frac{k(k+1)}{2} + (-1)^{k+2} (k+1)^2$

$= \frac{1}{2} (-1)^{k+1} (k+1) [k + (-1)^1 (k+1) \cdot 2]$

$= \frac{1}{2} (-1)^{k+1} (k+1) [- (k+2)]$

$= \frac{(k+1)(k+2)}{2} \cdot (-1)^{k+1} \cdot (-1)^1$

$= \frac{(k+1)(k+2)}{2} \cdot (-1)^{k+2}$

(1/2)

Question 3

(c) (i)

$$\text{let } A(x) = (x-\alpha)^2 Q(x) \quad \textcircled{\frac{1}{2}}$$

$$A'(x) = (x-\alpha)^2 Q'(x) + 2(x-\alpha)Q(x)$$

$$\Rightarrow A(\alpha) = (\alpha-\alpha)^2 Q(\alpha) = 0$$

$$A'(\alpha) = (\alpha-\alpha)^2 Q'(\alpha) + 2(\alpha-\alpha)Q(\alpha) = 0 \quad \textcircled{\frac{1}{2}}$$

$$A''(x) = (x-\alpha)^2 Q''(x) + 2(x-\alpha)Q'(x) + 2(x-\alpha)Q'(x) + 2Q(x)$$

$$A''(\alpha) = 0 + 0 + 0 + 2Q(\alpha) \quad \textcircled{\frac{1}{2}}$$

and $Q(\alpha) \neq 0$ [since double root]

$$\Rightarrow A''(\alpha) \neq 0 \quad \textcircled{\frac{1}{2}}$$

(ii) $A(x) = 5x^5 - 3x^3 + k$

$$A'(x) = 25x^4 - 9x^2 = 0 \quad \textcircled{1}$$

when $x=0$ or $x = \pm \frac{3}{5}$

$$A'(0) = 0 \text{ but } A(0) = k$$

$\therefore x=0$ not double root (unless $k=0$)

$$\left. \begin{array}{l} A'(\frac{3}{5}) = 0 \\ A''(x) = 100x^3 - 18x \\ A''(\frac{3}{5}) = 10.8 \end{array} \right\} \begin{array}{l} x = \frac{3}{5} \quad \textcircled{1} \\ \text{double root} \end{array}$$

$$\therefore A(\frac{3}{5})' = 5(\frac{3}{5})^5 - 3(\frac{3}{5})^3 + k = 0$$

$$\boxed{k = 0.2592}$$

~~Also A'~~

①

$$k = \frac{162}{625}$$

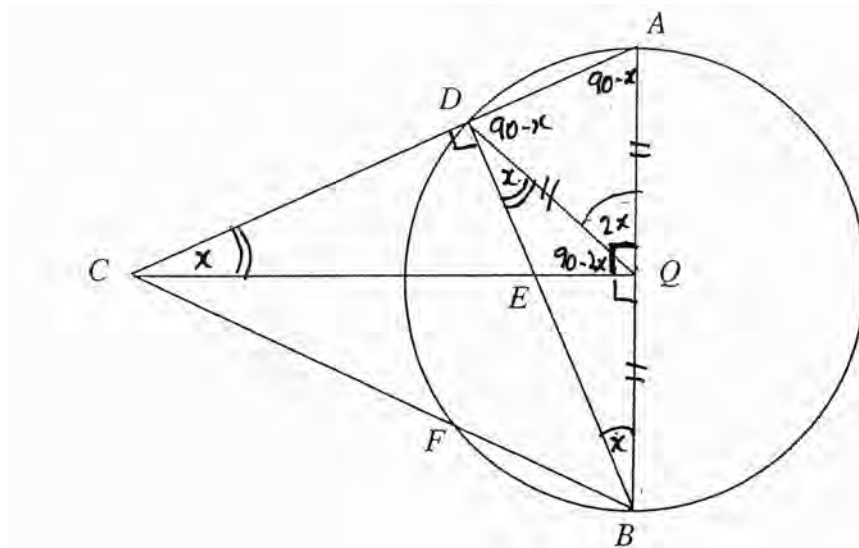
$$\begin{array}{r} 0.3888 - \\ 0.6480 \\ \hline 4.08 \end{array}$$

$$0.3888 -$$

$$\frac{3}{10} \rightarrow 45000$$

$$1300$$

- (d) AB is a diameter of the circle centre Q , $CQ \perp AB$ at Q , and AC and BC intersect the circle at D and F respectively.



- (i) Prove that $\angle QDB = \angle QCA$.
 (ii) Hence or otherwise show that quadrilateral $CDQB$ is cyclic.

This is the end of the paper.

(i) let $\angle QDB = x$ ✓
 $\angle QBE = x$ (isosceles $\triangle DQB$)
 $\angle A\hat{Q}D = 2 \times \angle A\hat{B}D$ (angle at centre th.)
 $\therefore \angle A\hat{Q}D = 2x$
 $\angle Q\hat{D}A = 90 - x$ (base angle $\triangle QDA$)
 $\therefore \angle Q\hat{C}A = x$ (angle sum $\triangle QCA$)

(ii) $\angle C\hat{D}E = 90^\circ$
 $(\angle A\hat{D}E + \angle C\hat{D}E = 180^\circ)$
 and $\angle A\hat{D}E = 90^\circ$
 $\angle C\hat{Q}B = 90^\circ$ (adj to $\angle C\hat{Q}A$ on str. line)
 $\therefore \angle C\hat{D}E = \angle C\hat{Q}B = 90^\circ$
 Both angles are subtended at cir. of circle through C, D, Q, B by same chord BC .
 (BC is the diameter of circle)